Reg No.:_______________ Name:__________________________

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Seventh Semester B.Tech Degree Regular and Supplementary Examination December 2023 (2019 Scheme)

Course Code: CST413

Course Name: MACHINE LEARNING

Max. Marks: 100 **Duration: 3 Hours**

PART A

Answer all questions, each carries 3 marks. Marks

1	What is machine learning? List out any three applications of machine learning.	(3)
2	Explain the general Maximum Likelihood Estimate method for estimating the parameters of a probability distribution.	(3)
3	Distinguish between classification and regression with suitable examples.	(3)
4	Discuss about the issues in decision tree learning and suggest methods to overcome it.	(3)
5	Calculate the output of the neuron y. The input feature vector is (x1, x2, x3) = (0.7,0.3,0.8) and weight values are [w1,w2,w3, b] =[0.2, 0.8, 0.2, 0.30]. Use sigmoid function as activation function.	(3)
6	Explain the significance of maximum margin hyperplane in SVM.	(3)
7	Calculate the similarity between two points A (13,15) and B (20,18) using: (i). Manhattan distance (ii) Euclidean distance (iii) Chessboard distance	(3)
8	Explain Expectation – Maximization algorithm.	(3)
9	Given 30 human photographs, a computer predicts 19 to be male (as positive class), 11 to be female. Among the 19 male predictions, 3 predictions are not correct. Among the 11 female predictions, 1 prediction is not correct. Compute the precision, recall and accuracy of the classifier.	(3)
10	Explain how bias-variance trade-off affects machine learning algorithms.	(3)

PART B

Answer any one full question from each module, each carries 14 marks.

Module I

11	a)	With necessary examples and diagrams, explain the concept of supervised, unsupervised and reinforcement learning.	(9)
	b)	State Bayes theorem and explain how Bayes theorem is used to compute the	(5)

probability P(c_k | X),where X is an instance of a feature vector and c_k is the class label.

OR

12 a) Consider the binomial distribution, which has p.m. f. (9)

$$f(x) = \binom{n}{x} p^x (1-p)^{n-x}$$

for x = 0,1, 2, . . . n.

i. Write an expression for the log-likelihood of the data as a function of the parameter p.

ii. Compute maximum likelihood estimate for p?

b) Differentiate between Maximum Likelihood estimation (MLE) and Maximum a Posteriori (MAP) estimation. (5)

Module II

13 a) Use the method of least squares to construct a linear regression model for the following data. Estimate the value of y when x=4. (5)

x	10	9	8	7	6	5
y	8	12	7	10	9	6

b) The following dataset shows the result of whether a person is pass or fail in the exam based on various outputs. Use Naive Bayes classifier to classify an **instance 'X' with <Confident=Yes, Studied=Yes, Sick=No>.** (9)

Confident	**Studied**	**Sick**	**Result**
Yes	No	No	Fail
Yes	No	Yes	Pass
No	Yes	Yes	Fail
No	Yes	No	Pass
Yes	Yes	Yes	Pass

OR

14 a) Identify the first splitting attribute for decision tree by using ID3 algorithm with the following dataset. (9)

Height	Hair	Eyes	Attractiveness
short	blond	blue	Yes
tall	blond	brown	No
tall	red	blue	Yes
short	dark	blue	No
tall	dark	blue	No
tall	blond	blue	Yes
tall	dark	brown	No
short	blond	brown	No

b) Explain different types of regularization and its significance. (5)

Module III

15 a) Describe perceptron training algorithm with neat diagram. What is the drawback of single layer perceptron? (7)

b) Explain how Support Vector Machine (SVM) can be used for classification of linearly separable data. (7)

OR

16 a) State the mathematical formulation of the SVM problem. Give an outline of the SVM Classifier Algorithm. (9)

b) Give the significance of kernel trick in the context of support vector machine. Show that the function $\boldsymbol{k(x,y)=(x.y)^3}$ is a kernel function where x.y represents dot product of x and y. (5)

Module IV

17 a) Distinguish between K-means clustering and hierarchical clustering. (5)

b) Suppose that the data mining task is to cluster the following eight points (with (x,y) representing location) into three clusters A1(2, 10), A2(2, 5), A3(8, 4), A4(5, 8), A5(7, 5), A6(6, 4), A7(1, 2), A8(4, 9).The distance function is Manhattan distance. Suppose initially we assign A1(2, 10), A4(5, 8) and A7(1, 2) as the centre for each cluster respectively. Use the K-means algorithm to find the three clusters and their centres after two rounds of execution. (9)

OR

18 a) Describe Principal Component Analysis algorithm used for dimensionality reduction. (5)

b) Construct dendrograms using single and complete linkage agglomerative clustering to group the data described by the following distance matrix. (9)

	A	B	C	D
A	0	1	4	5
B	1	0	2	6
C	4	2	0	3
D	5	6	3	0

Module V

19 a) Describe ROC space and ROC curve in machine learning. In ROC space, illustrate which points correspond to perfect prediction, always positive prediction and always negative prediction. Why? (7)

b) Differentiate between bagging and boosting. (7)

OR

20 a) Give the significance of cross validation technique and bootstrapping with neat diagrams. (7)

b) Describe the techniques used for ensemble learning. (7)

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